

Lab Report Problem of Advanced Statistical Inference

Solve using R or Python

Instructions: For each question, write code (R or Python) to load/simulate the data, compute the requested numerical quantities, and produce any requested plots. After each part, report your numbers in a short table and add 2–3 sentences of interpretation grounded in those numbers. You may use numpy/scipy/pandas/matplotlib in Python, or base R plus stats, MASS, and ggplot2 in R.

Submission Guidelines

- Submit a single R script/Markdown (.R/.Rmd) or Python script/Notebook (.py/.ipynb) containing all code, output tables, plots, and interpretations.
- Clearly label each part (Q1a, Q1b, ... Q6f).
- Set a random seed for every simulation (set.seed() in R, np.random.seed()/random.seed() in Python).
- Include all requested plots with proper axis labels and titles, and all requested numbers as tables.

Question 1

A machine fills bottles with a target volume $\mu = 10$ (mL) and known process variance $\sigma^2 = 4$. You will study several estimators purely through simulated filling data.

a) Bias check. Simulate $M = 5000$ independent samples of size $n = 20$ from $N(10, 4)$. For each sample, compute the sample mean $\bar{X}$, the sample variance with divisor $(n-1)$ [S^2], and the sample variance with divisor n [σ^2_{MLE}]. Average each estimator across the 5000 samples and put the three averages in a table next to the true values ($\mu = 10$, $\sigma^2 = 4$). State, purely from the numbers, which variance estimator is biased and in which direction.

b) Efficiency comparison. Using the same 5000 samples, compute $T_1 = \bar{X}$ and $T_2 = (X_1 + X_n)/2$ for each sample. Report the empirical variance of T_1 and T_2 in a table, and the ratio $\text{Var}(T_2)/\text{Var}(T_1)$. State which estimator is more efficient based on the ratio.

c) Sufficiency check. Generate two different samples of size $n = 20$ that have identical $\bar{X}$ and S^2 but different individual observations (e.g., swap/perturb pairs of values so the mean and variance are preserved). For each sample, compute the likelihood at 4–5 different (μ, σ^2) values and tabulate the likelihood ratio between the two samples at each parameter value. Report whether the ratio stays constant across parameter values.

d) Consistency check. For $n = 5, 10, 25, 50, 100, 250, 500, 1000$, simulate 2000 samples at each n and compute $\bar{X}$ for each. Produce one plot (boxplots or overlaid densities) showing the sampling distribution of $\bar{X}$ across n . Tabulate the empirical variance of $\bar{X}$ at each n alongside $4/n$, and tabulate the fraction of samples with $|\bar{X} - 10| > 0.1$ at each n .

(Optional add-on: redo part (d) using the sample median in place of the mean, and add its variance/fraction columns to the same table.)

Question 2

The following 15 recorded times (in years) until failure of a machine component are believed to follow an Exponential distribution:

2.1, 0.8, 3.4, 1.2, 5.6, 0.4, 2.9, 1.8, 4.1, 0.6, 3.3, 2.0, 1.5, 6.2, 0.9

a) Log-likelihood curve. Write code to evaluate and plot the log-likelihood $\ell(\lambda)$ for this data over a grid of λ from 0.05 to 2. Report the λ value on the grid that maximizes $\ell(\lambda)$.

b) Numerical MLE. Numerically maximize $\ell(\lambda)$ using an optimizer (optim() in R, or scipy.optimize.minimize_scalar / minimize in Python) and report $\hat{\lambda}$. Compare it to the grid-based maximum from part (a) and to $1/\text{mean}(\text{data})$.

c) Standard error and Wald CI. Compute the observed Fisher Information $I(\lambda)$ by numerically evaluating the second derivative of $\ell(\lambda)$ at λ (finite differences are fine). Use it to get the asymptotic standard error of λ and report a 95% Wald confidence interval.

d) Model comparison via AIC. Fit a Gamma distribution (shape k , rate θ) to the same 15 values by numerical MLE. Report the fitted parameters, the log-likelihood, and the AIC for both the Exponential and Gamma fits in a comparison table. State which model AIC prefers.

e) Visual fit check. Plot a histogram of the 15 data points with the fitted Exponential and Gamma density curves overlaid.

Question 3

A dataset of 20 sensor measurements is given below; a few values are believed to be contaminated outliers due to sensor error:

23.1, 24.5, 22.8, 25.0, 23.6, 24.1, 23.9, 22.7, 150.2, 24.8, 23.3, 24.0, 23.5, 22.9, -80.5, 24.2, 23.7, 24.6, 23.0, 24.3

a) Compare center estimates. Compute the sample mean, sample standard deviation, median, Median Absolute Deviation (MAD), and 10%-trimmed mean on the full (contaminated) dataset. Put all five numbers in a table. State which estimators are pulled the most by the two outliers.

b) Huber M-estimator. Compute a Huber M-estimate of location with tuning constant $k = 1.345$ (`MASS::rlm()/MASS::huber()` in R, or `statsmodels.robust.scale` / a short IRWLS loop in Python). Add it to the table from part (a).

c) Breakdown-point sweep. Programmatically increase the number of contaminated points from 0 to 9 (out of 20), recomputing the mean and median at each step. Plot both estimators against the number of contaminated points and report the contamination count at which the mean's value moves outside a reasonable range (e.g., more than 10 units from the clean-data mean).

d) Clean-data comparison. Remove the two obvious outliers and recompute the mean, median, MAD, trimmed mean, and Huber M-estimate on the remaining 18 points. Show a side-by-side table of the contaminated vs. clean-data results and summarize the differences in 3–4 sentences.

Question 4

A survey of 40 randomly selected voters asks whether they support a new policy. 27 out of 40 respond "Yes."

a) Compute the posterior numerically. Using a $\text{Beta}(2,2)$ prior for p , compute the posterior distribution on a fine grid of p values (e.g., 1000 points from 0 to 1) by multiplying the prior density by the $\text{Binomial}(40, p)$ likelihood at $x = 27$ and normalizing. Report that the resulting shape matches a Beta density and give its two parameters.

b) Plot all three curves. On one graph, plot the prior density, the (rescaled to a comparable height) likelihood, and the posterior density of p .

c) Point estimates. From the posterior grid, compute the posterior mean, posterior mode, and posterior median of p . Report these alongside the frequentist MLE $\hat{p} = 27/40$ in a single table.

d) Interval comparison. Compute a 95% credible interval for p from the posterior quantiles, and separately compute a 95% frequentist confidence interval (Wald or Wilson) from the same data. Report both intervals and their widths side by side.

e) Prior sensitivity table. Repeat parts (a)–(c) with two more priors: $\text{Beta}(1,1)$ and $\text{Beta}(20,20)$. Build one table with rows = $\{\text{Beta}(1,1), \text{Beta}(2,2), \text{Beta}(20,20)\}$ and columns = $\{\text{posterior mean, posterior mode, posterior median, 95\% credible interval}\}$. Comment on how much the estimates move across priors.

(Optional add-on: rebuild the same table assuming $n = 400$ voters with 270 "Yes" responses — same proportion, larger sample — and note how much less the priors move the estimates.)

Question 5

a) Two intervals for λ . Using the failure-time data from Question 2, compute a 95% confidence interval for λ using the formula based on $2n\lambda\bar{X} \sim \chi^2(2n)$, and also the Wald interval from Question 2(c). Report both intervals and their widths in a table.

b) Coverage simulation (t-based). Reusing the Question 1 simulation setup ($n = 20$, $\mu = 10$, $\sigma^2 = 4$, 5000 replications), construct a 95% t-based confidence interval for μ from each simulated sample. Compute and report the empirical coverage probability (fraction of the 5000 intervals containing $\mu = 10$).

c) Coverage simulation (z-misspecified). Repeat part (b) but use the normal quantile $z = 1.96$ instead of the correct t-quantile. Report empirical coverage for $n = 5$ and $n = 100$ side by side in a table, and state which sample size shows the bigger gap from 95%.

d) Bootstrap CI for the median. Using the contaminated dataset from Question 3, draw at least 2000 bootstrap resamples and construct a 95% percentile bootstrap confidence interval for the median. Report this interval alongside a naive normal-theory interval for the mean computed on the same contaminated data.

Question 6

A production line's defect rate is tested sequentially: $p_0 = 0.05$ (acceptable) vs. $p_1 = 0.15$ (unacceptable), inspecting items one at a time ($X_i = 1$ if defective, else 0).

a) Boundaries. With $\alpha = 0.05$ and $\beta = 0.10$, compute the SPRT boundaries $A \approx (1-\beta)/\alpha$ and $B \approx \beta/(1-\alpha)$, and report $\log A$ and $\log B$.

b) Implement the update rule. Using the update $\log \Lambda_n = \log \Lambda_{n-1} + X_n \cdot \log(p_1/p_0) + (1-X_n) \cdot \log((1-p_1)/(1-p_0))$, write code that takes a stream of Bernoulli observations, updates $\log \Lambda_n$ one observation at a time, and stops the first time the running value crosses $\log B$ (accept H_0) or $\log A$ (reject H_0).

c) Simulation study — true rate p_0 . Simulate 1000 independent SPRT runs with true defect rate $p = 0.05$. Report the proportion of runs that correctly accept H_0 , and the mean and median stopping sample size, in a table.

d) Simulation study — true rate p_1 . Repeat part (c) with true defect rate $p = 0.15$. Report the proportion of runs that correctly reject H_0 , and the mean stopping sample size.

e) Fixed-sample comparison. Using the normal-approximation sample-size formula for a one-sample proportion test, compute the fixed sample size n needed for $\alpha = 0.05$ and power = 0.90 to distinguish $p_0 = 0.05$ from $p_1 = 0.15$. Report this n next to the average stopping sizes from parts (c) and (d) in a single table.

f) Sample-path plot. Plot 5–10 example sample paths of $\log \Lambda_n$ against n for the $p = 0.05$ scenario and 5–10 for the $p = 0.15$ scenario (two panels), with horizontal reference lines at $\log A$ and $\log B$, showing where each path terminates.